

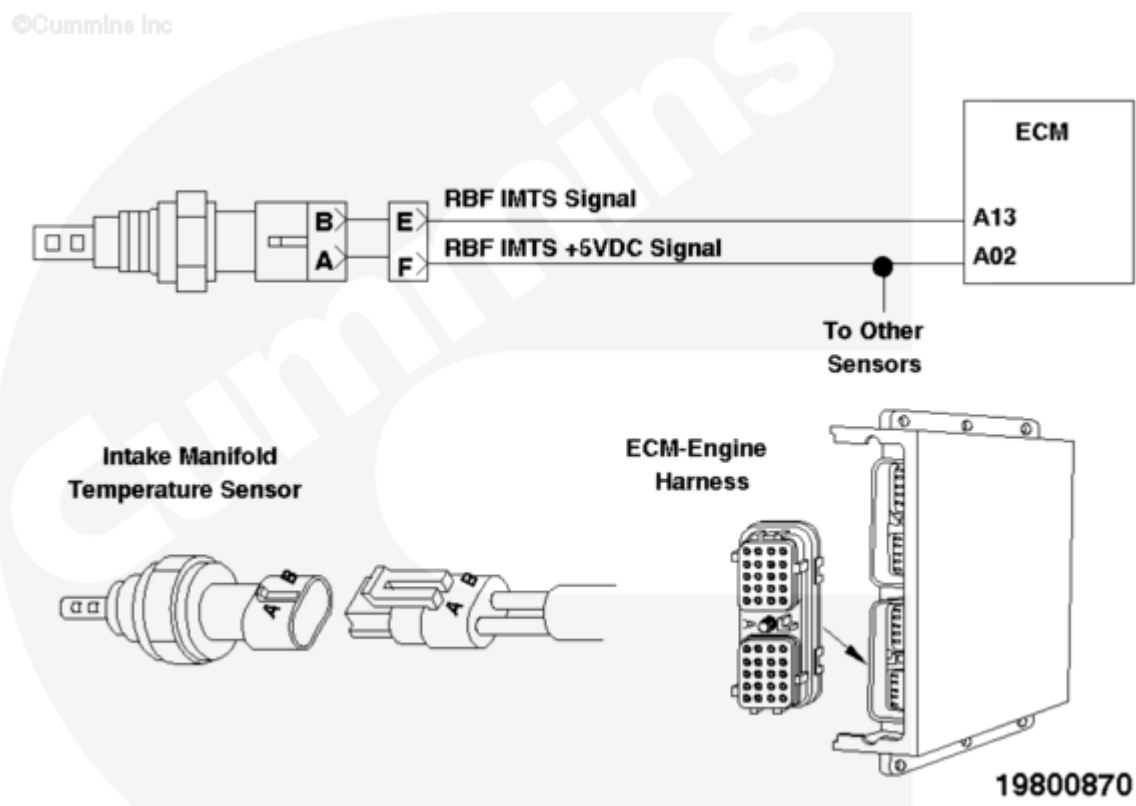
## Fault Code: 159

### Intake Manifold 3 Temperature Sensor Circuit - Voltage Above Normal or Shorted to High Source

 Printable Version

#### Overview

| Codes                                                                                    | Reason                                                                                                                                                                                       | Effect                      |
|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| Fault Code: 159<br><br>PID(P): 105<br>SPN: 1132<br>FMI: 3<br>Lamp: Yellow<br>SRT: 00-698 | Intake Manifold 3<br>Temperature Sensor Circuit -<br>Voltage Above Normal or<br>Shorted to High Source.<br>High voltage detected on<br>SIGNAL pin 13 of the main<br>harness A ECM connector. | Fault Code 162 is disabled. |



Right Bank Front Intake Manifold Temperature Sensor Circuit

#### Circuit Description

The right bank front intake manifold temperature sensor circuit provides the right bank front intake air temperature signal to the ECM. The resistance of the sensor varies with temperature. The ECM detects the change in resistance of the sensor by monitoring the voltage across an internal resistor that is in series with the sensor. The change in voltage across the internal resistor is translated into a temperature change.

## Component Location

The right bank front intake manifold temperature sensor is located on the right bank front intake manifold. The sensor is located downstream of the aftercooler.

## Shoptalk

- The resistance of the sensor varies with the temperature. The reading that you observe should compare to the following table if the sensor is functioning properly.

| Temperature | Temperature | Resistance   |
|-------------|-------------|--------------|
| (°C)        | (°F)        | (ohms)       |
| 0           | 32          | 30k to 36k   |
| 25          | 77          | 9k to 11k    |
| 50          | 122         | 3k to 4k     |
| 75          | 167         | 1350 to 1500 |
| 100         | 212         | 600 to 675   |

Refer to Troubleshooting Fault Code t05-159 (</qs3/pubsys2/xml/en/procedures/81/81-t05-159.html>)